



Solar-Powered Adaptive Traffic Light System

Smart Traffic Control Using Vehicle Density Detection

PROBLEM STATEMENT:

- *Traditional traffic lights run on fixed timers.*
- *Vehicles may wait even when roads are empty.*
- *Leads to traffic congestion and wasted travel time.*
- *Increased fuel consumption & pollution.*
- *Signals do not adapt to changing traffic flow*



SOLUTION

- ◆ *A solar-powered, adaptive traffic light system.*
- ◆ *Automatically adjusts green time based on traffic density*
- ◆ *Eco-friendly & cost-effective, suitable for smart cities.*
- ◆ *Uses IR sensors or image processing to count vehicles.*
- ◆ *Reduces waiting time, fuel usage, and congestion.*

FEATURES

- Solar-powered using PV panels with rechargeable batteries for 24/7 operation.
- Real-time traffic detection via IR, ultrasonic, or AI cameras.
- Microcontroller-based adaptive signal timing for smooth flow.
- Reduces congestion, fuel use, and waiting time.
- Emergency vehicle priority with remote green signal control.
- Energy-efficient LED lights for visibility and durability.
- Standalone system requiring minimal infrastructure.
- Eco-friendly & cost-effective for smart urban and rural traffic management.



COMPONENTS

Microcontroller (Arduino Uno)

Using input from IR sensors, it processes vehicle data to make timing decisions and control the traffic light outputs in the correct sequence.

Ultra Sonic Distnce Sensor

detects reflected light from vehicles to count waiting cars in each lane

LED

The red, yellow, and green LEDs serve as the traffic lights, signaling stop, caution, and go.

Solar Panel

A 6V solar panel converts sunlight into electricity to charge the battery. This provides a renewable power source, demonstrating the project's green energy concept.

COMPONENTS

18650
Lithium-ion
Battery

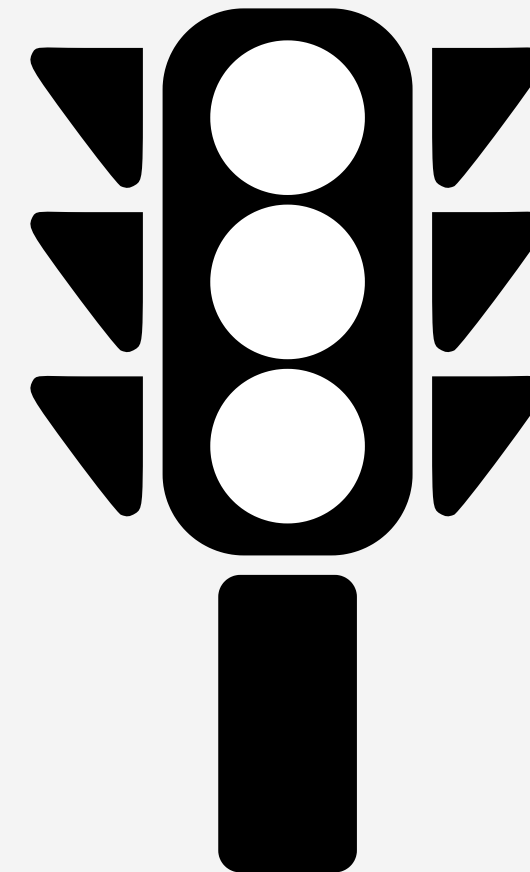
Provides a stable, independent power supply during nighttime or cloudy conditions

TP4056
Charging
Module

This smart charger safely manages the battery charging process from the solar panel, preventing overcharging and providing clear status indicators to ensure battery health and longevity.

Voltage
Regulator

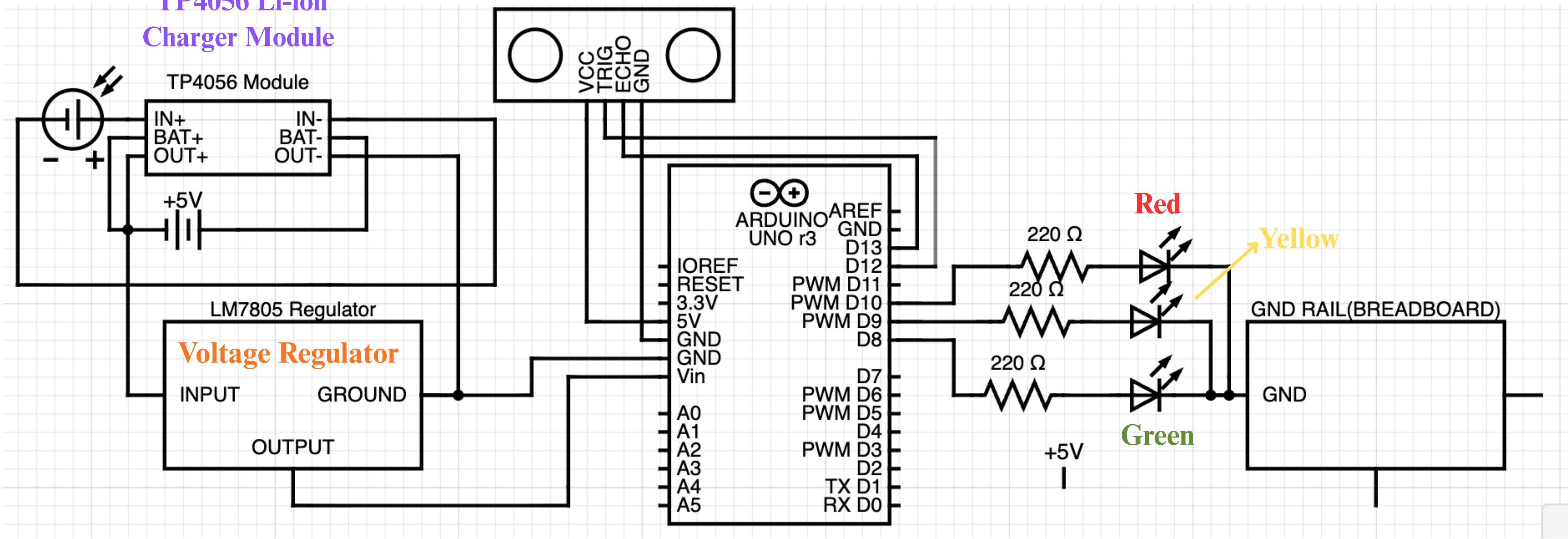
maintains a constant voltage level regardless of changes in input voltage or load current.



CIRCUIT DIAGRAM

Charging Circuit:
TP4056 Li-ion
Charger Module

Ultrasonic distance sensor-Car Sensor



ARDUINO CODE

```
// Define Pins
const int trigPin = 12;
const int echoPin = 13;
const int redLight = 8;
const int yellowLight = 9;
const int greenLight = 10;

// Variables for counting and timing
int carCount = 0;
const int triggerDistance = 20; // Distance in cm to detect a car
const int maxCars = 40;
const unsigned long greenInterval = 30000; // 30 seconds green time
bool carDetected = false;

// Timer variables
unsigned long previousMillis = 0;
bool greenLightActive = false;
unsigned long greenStartTime = 0;

void setup() {
  // Set pin modes
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(redLight, OUTPUT);
  pinMode(yellowLight, OUTPUT);
  pinMode(greenLight, OUTPUT);

  // Start with Red Light
  digitalWrite(redLight, HIGH);
  digitalWrite(greenLight, LOW);
  digitalWrite(yellowLight, LOW);

  // For debugging
  Serial.begin(9600);
  Serial.println("Traffic Light Started - Less than 40 cars = Green Light");
}
```

```
void loop() {
  unsigned long currentMillis = millis();

  // Measure distance from ultrasonic sensor
  long duration, distance;
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  duration = pulseIn(echoPin, HIGH);
  distance = (duration / 2) / 29.1; // Convert to cm

  // Car detection logic
  if (distance < triggerDistance && !carDetected) {
    carDetected = true;
    carCount++;
    Serial.print("Car Detected! Total Cars: ");
    Serial.println(carCount);
  }

  // Reset car detection when car leaves sensor area
  if (distance >= triggerDistance && carDetected) {
    carDetected = false;
  }

  // MAIN LOGIC: Control traffic light based on car count
  if (carCount < maxCars) {
    // LESS THAN 40 CARS - Use interval-based green light

    if (!greenLightActive) {
      // Start green light cycle
      startGreenCycle();
    } else {
      // Green light is active - check if interval is over
      if (currentMillis - greenStartTime >= greenInterval) {
```

```

endGreenCycle();
}
}

} else {
// 40 OR MORE CARS - Turn light red
if (greenLightActive) {
endGreenCycle();
}
digitalWrite(redLight, HIGH);
digitalWrite(yellowLight, LOW);
digitalWrite(greenLight, LOW);

// Optional: Reset counter after red period or keep it red
Serial.println("40 cars reached! Light remains RED.");

// Uncomment below to auto-reset after delay:
// delay(60000); // Red for 60 seconds
// carCount = 0; // Reset counter
// Serial.println("Counter reset to 0");
}

delay(100); // Short delay between readings
}

void startGreenCycle() {
greenLightActive = true;
greenStartTime = millis();

// Green light sequence
digitalWrite(yellowLight, LOW);
digitalWrite(redLight, LOW);
digitalWrite(greenLight, HIGH);

Serial.println("Green Light ACTIVATED - Interval started");
Serial.print("Current car count: ");
Serial.println(carCount);
}

```

```

Serial.print("Current car count: ");
Serial.println(carCount);
}

void endGreenCycle() {
// Yellow warning before red
digitalWrite(greenLight, LOW);
digitalWrite(yellowLight, HIGH);
Serial.println("Yellow Light - Preparing for Red");
delay(3000); // Yellow for 3 seconds

// Turn red
digitalWrite(yellowLight, LOW);
digitalWrite(redLight, HIGH);
greenLightActive = false;

Serial.println("Green cycle ended - Red Light ACTIVATED");

// Optional: Small delay before next cycle can start
delay(2000);
}

```

MATERIAL COST

Component	Quantity	Unit Price (₹)	Total Price (₹)
Microcontroller			
Arduino Nano (Clone)	1	₹279	₹279
Power System			
6V 1W Solar Panel	1	₹189	₹189
18650 3.7V Battery	1	₹199	₹199
TP4056 Charger Module	1	₹89	₹89
LM7805 Voltage Regulator	1	₹24 (pack of 5)	₹24
Sensors & Output			
HC-SR04 Ultrasonic Sensor	1	₹99	₹99
5mm LED Pack (10 pieces)	1	₹45	₹45
220Ω Resistor Pack (50 pieces)	1	₹29	₹29
Accessories			
400 Points Breadboard	1	₹129	₹129
Jumper Wires (40 pieces)	1	₹69	₹69
TOTALS	9 Items		₹951

ROLES

ROHAN MENON: *Computer Science Engineering (CSE)*

Software integration and algorithm design — developing the main control logic, integrating sensor inputs, and ensuring smooth communication between modules.

SWETHA SENTHIL KUMAR SINI *Computer Science Engineering (CSE)*

Data flow management and user interface development — handling code optimization, user interaction logic, and ensuring the system's firmware is modular and scalable.

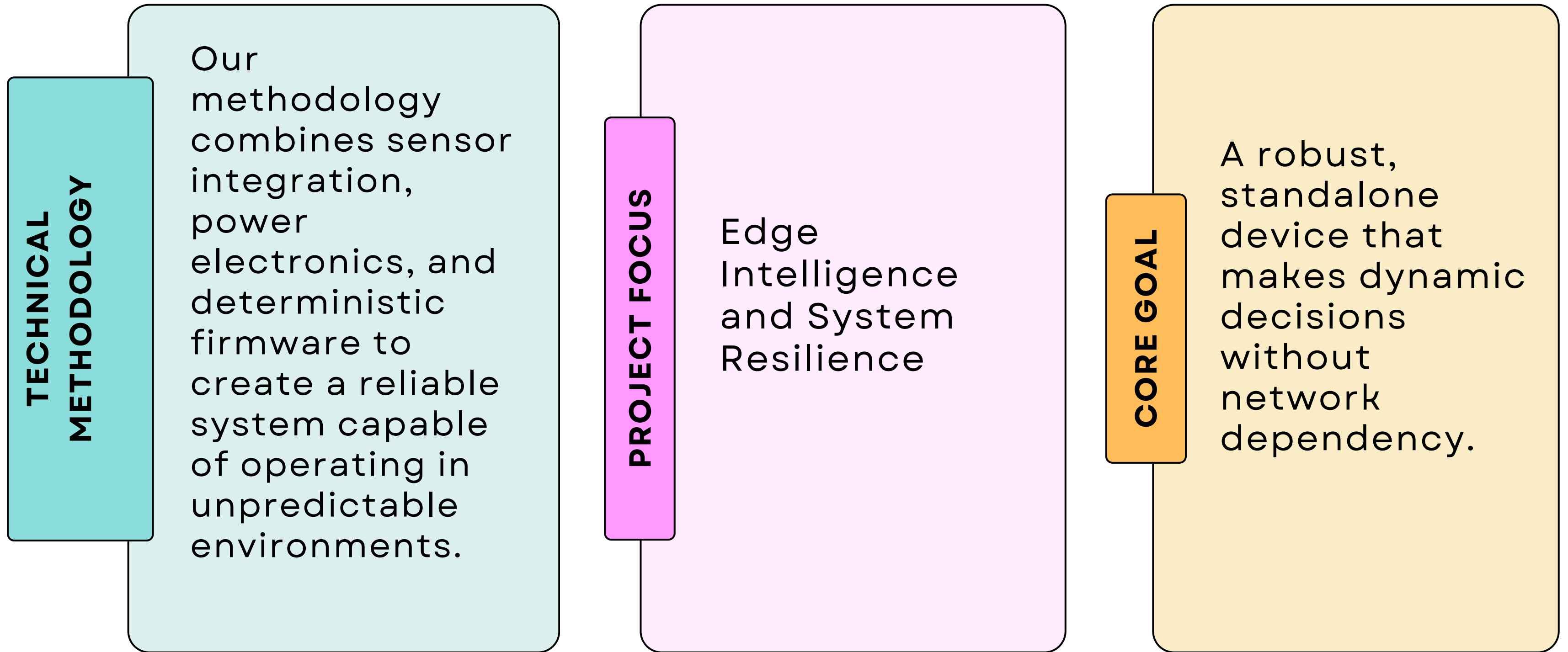
SHIVA S *Computer Science Engineering (CSE)*

Embedded code deployment and database/log management — maintaining code version control, handling data storage if applicable, and ensuring proper integration between software and hardware components.

YUVAN SHANKAR *Btech EE(VLSI)*

Low-power circuit design, sensor integration, and firmware development for standalone IoT devices.

TECHNICAL FRAMEWORK



Thank You!